

An Introduction to IoT Constrained Node Networks

In this first part of a series on IoT constrained node networks, we get introduced to the IoT, its building blocks, constrained node networks, their characteristics, the protocols used and the recommended firmware.

The Internet of Things (IoT) transforms our lives through smart automation and has become an integral part of many activities. IoT is spreading to various domains such as industry, agriculture, medicine, smart cities, etc. The main elements that have enabled widespread IoT adoption are the rapid advances in software, networking, sensors and artificial intelligence (AI). IoT involves giving networking capabilities to things, which can be physical or virtual. Physical things include humans, wearables, vehicles, homes, computers,

embedded devices, etc. Virtual things include virtual machines, virtual networks, social media sites, etc. All these things being connected through the Internet is what IoT is all about.

The building blocks of an IoT system

An end-to-end IoT system can be classified into three networks — the edge network, the fog network and the cloud network. This classification is based on the type of devices and the applications running in those networks.

A. Edge network

- *Devices:* These include embedded devices, sensors, actuators and gateways. The devices may be constrained (with limited resources) or unconstrained.
- *Applications:* These include sensing, monitoring, actuating and conducting operations with the external world.

B. Fog network

- *Devices:* These are gateways and high-end servers.
- *Applications:* These include the roles of the broker, data acquisition and processing, commanding, real-time analytics, databases, etc.

C. Cloud network

- *Devices:* These are cloud platforms and high-end servers.
 - *Applications:* These include storage, machine learning and deep learning.
- The edge system consists of constrained or unconstrained devices, or both. This article discusses only constrained devices.

Devices with limited resources like memory, processing capacity and power are called constrained nodes.

There are three classes of constrained nodes, as shown in Table 1.

Class	RAM	Flash
Class 0	< 10KiB	< 100KiB
Class 1	~ 10KiB	~ 100KiB
Class 2	~ 50KiB	~ 250KiB

Table 1: Classes of constrained nodes

The support for the network stack and the security for these constrained nodes is shown in Table 2.

Constrained networks

A constrained network exhibits the following characteristics:

- Low bit-rate/throughput
- High packet loss and high variability of packet loss
- Highly asymmetric link characteristics
- Lack of advanced network services like multi-cast

The possible reasons for these characteristics of constrained networks are listed below:

- Cost constraints
- Constraints posed by the nodes
- Physical constraints, e.g., power or environmental constraints
- Technology constraints such as having to use legacy technologies

Constrained node networks

A constrained network is composed of a significant number of constrained nodes. These constrained node

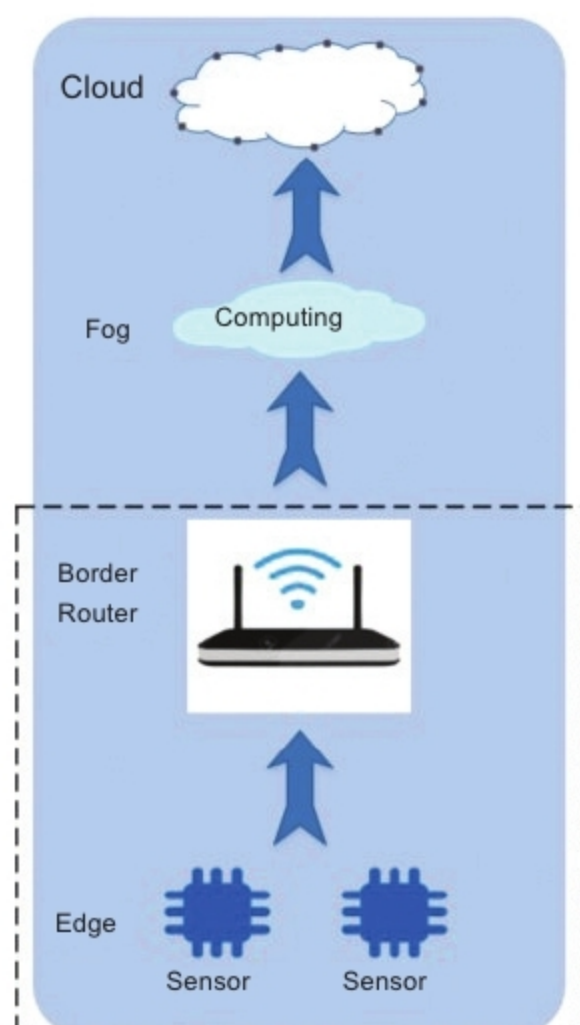


Figure 1: IoT building blocks show this network classification